

Option B · automatic refrigerated milk

Compared with the same coffee-only baseline. In addition to milk drinks, keeping milk ready between sessions introduces cold storage and automatic metering. Cleaning and space requirements gain new interfaces.

Existing needs + milk convenience → Functions		Weight						
		Prepare coffee	~ Clean coffee and milk paths	~ Contain coffee and steam hazards	+ Heat milk	+ Texture milk	+ Store refrigerated milk	+ Meter milk automatically
Enjoy good coffee	10	●	○		△ ⁺	△ ⁺	△ ⁺	△ ⁺
Clean up easily	7	○	●	△	○ ⁺	○ ⁺	● ⁺	● ⁺
Use little counter space	4	○	○	△	○ ⁺	○ ⁺	● ⁺	○ ⁺
Use safely at home	10	○	○	●	● ⁺	● ⁺	○ ⁺	○ ⁺
+ Enjoy warm, textured milk	8	△ ⁺	○ ⁺	○ ⁺	● ⁺	● ⁺	○ ⁺	● ⁺
+ Keep milk ready between sessions	6		● ⁺	○ ⁺			● ⁺	○ ⁺
Σ abs		161	213	143	205	205	217	205
Rel. %		12	16	11	15	15	16	15

Relation

● Strong (9)
○ Medium (3)
△ Weak (1)

Changes

Added
Removed
Changed

Option B • component consequences

A refrigerated reservoir, milk circuit, and automatic mixer require coordinated controls, cleaning access, and power/protection. Compare this architecture with the manual option; these are alternative concepts, not measured performance rankings.

Functions • inherited priorities → Components		Rel. %	Coffee brewing module	~ Controller with automatic milk cycle	~ Housing and shared power protection	+ Steam generator	+ Automatic steam–milk mixer	+ Refrigerated reservoir	+ Milk pump, valves, and tubing
~ Prepare coffee	11.9	●	○	○					
~ Clean coffee and milk paths	15.8	●	● 3→9	△	△ ⁺	● ⁺	● ⁺	● ⁺	● ⁺
~ Contain coffee and steam hazards	10.6	○	● 3→9	●	● ⁺	● ⁺	○ ⁺	○ ⁺	
+ Heat milk	15.2	△ ⁺	● ⁺	○ ⁺	● ⁺	● ⁺			○ ⁺
+ Texture milk	15.2		● ⁺	○ ⁺	● ⁺	● ⁺			○ ⁺
+ Store refrigerated milk	16.1		○ ⁺	○ ⁺			● ⁺	△ ⁺	
+ Meter milk automatically	15.2		● ⁺	△ ⁺		○ ⁺	○ ⁺	● ⁺	
Σ abs		296.5	731.9	301.6	384.7	556.6	364.3	417.9	
Rel. %		10	24	10	13	18	12	14	

Relation
● Strong (9)
○ Medium (3)
△ Weak (1)
Changes
Added
Removed
Changed